Bernard Balko

Mechatronics & Robotics Engineer

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SUMMARY

Robotics engineer with 3+ years of mechanical design experience at Pratt & Whitney and a current M.S. in Mechatronics & Robotics at NYU Tandon. Specializes in kinematics, embedded systems, and sensor fusion — bridging precision mechanical design with software and control. Published ASME researcher in autonomous aerial vehicle mapping.

EDUCATION

New York University, Tandon School of Engineering — 2026 – Present

M.S. Mechatronics & Robotics (Currently Enrolled)

Coursework: Foundations of Robotics, Mechatronics, Embedded Systems, Control Systems, Sensor Fusion

University of Hartford — May 2022

B.S. Mechanical Engineering | *Minor in Mathematics*

Dean's List: Spring 2019 – Spring 2021 • Senior Capstone published at ASME IMECE 2022

TECHNICAL SKILLS

Robotics & Control: Forward/Inverse Kinematics, Jacobian, DH Parameters, PD Control, Trajectory Tracking, Sensor Fusion, MuJoCo

Embedded Systems: STM32, Arduino, Embedded C, I2C/ADC, IMU (MPU6050), Haptic/LED Feedback, Signal Filtering

Programming: Python, C++, C, MATLAB, scipy

3D Design & CAD: Siemens NX, SolidWorks, OnShape, AutoCAD, GD&T, DFA, 3D Printing

Simulation & Analysis: MuJoCo, COMSOL, ANSYS Workbench, FEA, Root Cause Analysis

Perception & Vision: LiDAR, ICP, MSAC/RANSAC, Point Cloud Processing, Voxelization, Machine Vision, MATLAB

PROJECTS

Smart Posture Trainer (Wearable Mechatronic System) | — 2026

- Designed a real-time wearable posture monitor using an MPU6050 IMU and flex sensor on a chest strap controlled by an STM32H503RB microcontroller
- Implemented dual-sensor fusion via I2C/ADC with low-pass filtering and calibration-based baseline to eliminate false positives from natural movement
- Triggered haptic (vibration motor) and visual (LED) feedback only when poor posture persisted beyond 1 second — demonstrating embedded control logic and interrupt-driven design

UR10e 6-DOF Robot Kinematics — MuJoCo Simulation | — 2026

- Implemented full forward and inverse kinematics (position & velocity) from scratch for the UR10e manipulator inside a MuJoCo physics simulation
- Derived homogeneous transformation matrices from DH parameters; solved nonlinear IK using scipy optimization
- Computed the Jacobian for velocity-level kinematics and implemented a PD torque controller achieving stable joint-angle tracking

Aerial Vehicle Rapid 3D Map Generation — ASME Published | — 2022

- Designed a LiDAR scanning turret on a pan-tilt mechanism with IMU feedback for autonomous safe-landing zone detection in UAVs
- Applied ICP point cloud stitching and MSAC plane segmentation to classify flat, wall, and non-flat terrain surfaces
- Benchmarked 1×1, 5×5, and 10×10 voxelization grids for optimal speed/accuracy trade-off; results published at ASME IMECE 2022

Car Lane Detection System (Machine Vision) | 2021

- Built a MATLAB-based machine vision system to identify traffic lanes from a given video feed, keeping the vehicle centered within detected lane boundaries

WORK EXPERIENCE

Mechanical Design Engineer | *Quest Global — Pratt & Whitney (Remote)* June 2022 – Dec 2025

- Led a team of 3 engineers on F135/F119 engine inspection documentation, improving process efficiency by 30% and doubling project scope
- Conducted GD&T analysis on F135 STOVL 3BSM engine components, identifying risks and developing mitigation strategies to improve product reliability
- Performed root cause analysis on nonconforming PW4000-100, PW4000-94, PW2000, and F117 parts; implemented corrective actions and cross-functional process improvements
- Developed custom Excel tools for bearing shear stress and tolerance stack-up calculations, increasing precision in part assessments
- Used Siemens NX for 3D modeling and structural analysis; collaborated with structural, manufacturing, and aero engineering teams

PUBLICATIONS

Tatoglu A., Yin C.C., Cervello B., Corrado A., Balko B., Sohn K. *"Aerial Vehicle Rapid 3D Map Generation for Safe Landing Area Detection."* ASME IMECE 2022, Paper no. IMECE2022-95849. Columbus, Ohio, Oct–Nov 2022.

LANGUAGES

English (Native) • Korean (Proficient)